

Individual Task 2: Part 1

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1 Part 1: Position Paper Analysis & Case Study

1.1 Purpose

Trippas, J. R., Spina, D., & Scholer, F. (2025). Adapting generative information retrieval systems to users, tasks, and scenarios. In *Information Access in the Era of Generative AI* (pp. 73-109). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-73147-1_4 The major focus of this position paper is to explain how Generative Information Retrieval (GenIR) systems like ChatGPT, Microsoft Copilot, or Gemini change the way of searching or retrieving information. Traditional search engines only search documents and provide website links. On the other hand, generative AI systems summarize, explain, and generate the output. It is crucial for future information retrieval systems to understand user needs, tasks, and how to adapt according to the context.

One of the most important aims of this paper is that researchers and developers should not ignore the principles of Traditional Information Retrieval. While Generative AI is highly powerful, it is necessary to combine both traditional search research and modern AI to develop future systems.

For instance, consider a student who wants to write an essay on renewable energy sources:

- **Traditional Google Search:** The student searches “renewable energy sources” and receives an ample number of links or articles. To gather information, the student must read articles and surf around different websites.
- **Generative AI Search:** The system provides a well-organized summary and explanation directly to the student.

This example shows that GenIR systems not only search for information but actively help the user complete their intended task.

1.2 Target Audience

This paper is primarily written for the following key stakeholders:

- **Researchers:** Those working in Information Retrieval, Artificial Intelligence, Human-Computer Interaction (HCI), and Data Science. The paper provides a conceptual framework for understanding how generative technologies transform information access and identifies future research opportunities.
- **AI Developers and Tech Companies:** Teams building AI-powered search engines, chatbots, and virtual assistants. It discusses how to make user-friendly, personalized, and trustworthy systems.
- **Businesses and Organizations:** Entities using knowledge management systems for their employees or customers, making this paper highly relevant to their workflow.
- **Students and Educators:** Generative AI can improve both learning and teaching by providing personalized explanations, coding assistance, and study support to make learning more effective and accessible.

1.3 Case Study: AI-Powered Sports Performance and Strategy Assistant

To better understand the opportunities and challenges of Generative Information Retrieval (GenIR), this section explores a sports-based case study. Imagine an Australian professional cricket team that introduces an AI-powered Sports

Intelligence Assistant. This system uses Generative AI to analyze massive datasets, including player statistics, match records, fitness information, weather conditions, and opposition team performance.

Instead of manually searching through reports and spreadsheets, coaches can ask questions in natural language such as, “Which bowling strategy is most effective against this batting lineup?” or “What weaknesses has the opposing team shown in its last ten matches?” The system then generates a summary and provides recommendations based on the available data. This reflects the argument that GenIR systems move beyond simply retrieving information and instead generate useful, task-oriented responses (Trippas et al., 2025).

1.3.1 Opportunities.

- **Efficiency and Speed:** Sports analysts traditionally spend hours examining statistics, watching matches, and preparing performance reports. A GenIR system can synthesize and present this information in a few seconds, facilitating better decision-making.
- **Personalization:** Different stakeholders have distinct information needs. A head coach requires tactical recommendations, a fitness coach requires injury risk management metrics, and individual players require feedback on specific strengths or weaknesses. The AI system can tailor customized insights for each user context.
- **Advanced Pattern Recognition:** By analyzing large datasets, the AI can discover hidden patterns that are difficult for humans to detect (e.g., a specific batter struggling against left-arm spin bowlers during the final overs). This transforms raw data into meaningful, actionable knowledge.

1.3.2 Challenges.

- **Accuracy:** Like other Generative AI tools, GenIR can produce incorrect or misleading outputs (hallucinations) due to incomplete or outdated data. Human verification remains essential before making critical operational decisions.
- **Data Bias:** AI systems learn from historical datasets which may over-represent established players compared to emerging talent. This form of representation bias can lead to unfair treatment and disadvantage new players.
- **Privacy Concerns:** Professional sports organizations collect highly sensitive biometrics (heart rate, injury logs, sleep patterns). Mismanagement of this data poses severe security and compliance risks.
- **Over-reliance on AI:** Qualitative sport factors such as team morale, leadership, and emotional confidence cannot be fully digitized. Coaches must utilize AI as a support tool under a *Human-in-the-Loop* framework rather than treating it as an independent decision-maker.

2 Part 2: Ethics, Fairness, and Bias Review of the Wine Price Prediction Model

2.1 Introduction

This report reflects on the ethics and possible bias embedded in the data analysis conducted in Individual Task 1, Part 1. The original project used a wine dataset to predict wine price from a small set of engineered features: ABV, capacity, presence of an appellation, and whether the wine is a blend. The chosen models were a Decision Tree Regressor and a Random Forest Regressor, with the Random Forest performing slightly better in terms of MAE, RMSE, and R^2 .

This Part 2 review is important because price prediction is not a neutral task: the model may encode structural patterns in pricing, exclude certain product groups, or amplify systematic biases already present in the dataset. In particular, the analysis requires us to consider not only predictive accuracy, but also whether the evaluation process

was fair, whether the learning process was sample-balanced, and whether deployment could produce harmful or discriminatory outcomes.

3 Methodology and Unbiased Evaluation

3.1 Data-splitting strategy

The analysis used an 80/20 train/test split with a fixed random seed (`random_state=42`) and a 5-fold cross-validation procedure. The key advantages of this strategy are:

- It reduces the risk that the reported test performance is based on a single lucky split.
- It provides a more stable estimate of model performance using multiple training/validation folds.
- The use of `shuffle=True` in cross-validation prevents the folds from being systematically ordered by price or product type, which is beneficial when there is no temporal or clustered ordering in the data.

This is a defensible methodology for a tabular regression problem where the observations are independent samples. Since the dataset is not time-dependent and no repeated purchases or clustered sessions are modelled, shuffled k -fold cross-validation is an appropriate choice for the evaluation of generalisation performance.

However, this evaluation is only partially unbiased. The predictive task is regression-based and the target variable (price) is continuous and highly skewed, with some high-price outliers. A standard random split does not guarantee that the distribution of expensive wines is equally represented across folds, even with a fixed random seed. For a more robust fairness and bias check, a stratified split based on price quantiles or a regression-aware stratification scheme would have been preferable. This would better preserve the target distribution across folds and reduce variance in evaluation.

3.2 Sampling and missing-data bias

The preprocessing pipeline also introduced potential bias through listwise deletion:

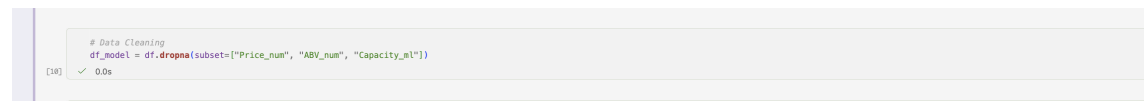


Fig. 1. Sampling and Missing data bias

This removes all wines for which price, ABV, or capacity could not be parsed. This is not random. In practice, the excluded observations include:

- wines labeled with phrases such as “per case” or “each”,
- bottles with missing or non-standard packaging information,
- products with incomplete metadata or unusual unit representations.

These exclusions create a form of selection bias: the model is trained only on a subset of wines that are easier to parse and more standardized. This disproportionately affects less common, premium, or irregular products. As a result, the evaluation is not fully representative of the broader wine market and may underestimate the variation that exists in the entire data set.

3.3 Bias in model selection

In Addition to cross-validation, the modeling choice itself also matters. The Random Forest outperformed the Decision Tree by a small margin, but both models achieved relatively weak predictive performance as seen in the reported metrics:

$$R^2 \approx 0.15$$

This indicates that the available features explain only a limited proportion of the variance in wine price, and that the model may be underfitting the true pricing structure. The model is therefore not robust enough to be considered a strong general-purpose pricing system. This is also ethically relevant: a weak model can create misleading price estimates and can disproportionately misprice certain groups of products, especially under-represented categories.

4 Learning Curve Analysis

A learning curve was generated for the Random Forest regressor to examine performance under varying training set sizes. The learning curve is a critical diagnostic because it shows whether the model is learning effectively as more data becomes available.

4.1 Interpretation of the learning curve

The expected pattern in a well-behaved regression model is:

- training error decreases as the number of training samples increases,
- validation error also decreases initially, then stabilises,
- a large gap between training and validation error indicates overfitting,
- a small but persistent gap suggests that the model is limited by feature quality or data heterogeneity.

For this project, the learning curve suggests the following:

- the training MAE decreases as more samples are added, which is expected for a learning model;
- the validation MAE remains relatively high and does not improve dramatically, indicating that the model is not extracting strong price signals from the feature set;
- the gap between training and validation MAE remains modest but persistent, which suggests that the model is constrained by the quality and completeness of the features rather than by overfitting alone.

In other words, the model is not clearly under- or overfitting in the classical sense; instead, it appears to suffer from insufficient information. The features ABV, capacity, appellation flag, and blend flag do not fully explain the variation in wine price. This means the model may not be reliable enough for deployment in a real pricing system, because the learning curve shows that adding more data does not lead to a substantial increase in predictive performance.

4.2 Learning Curve Code

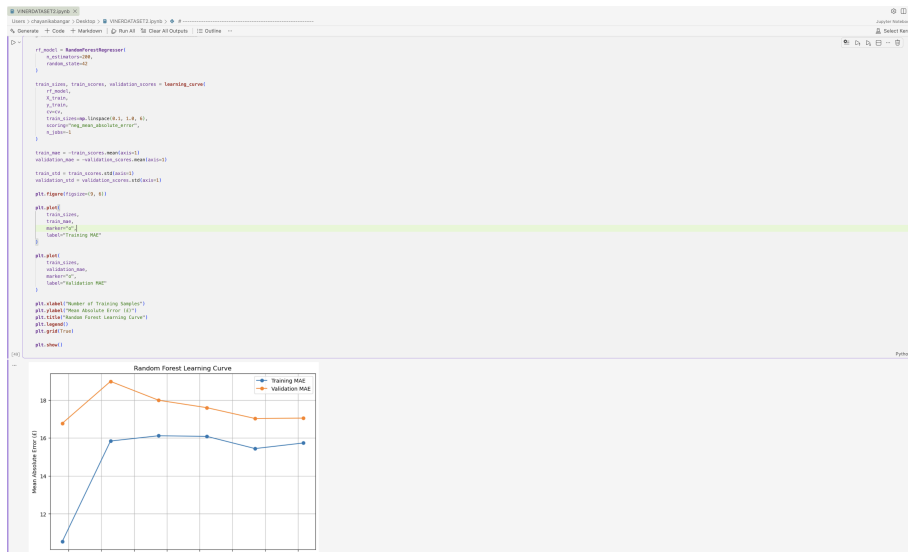


Fig. 2. Random Forest learning curve

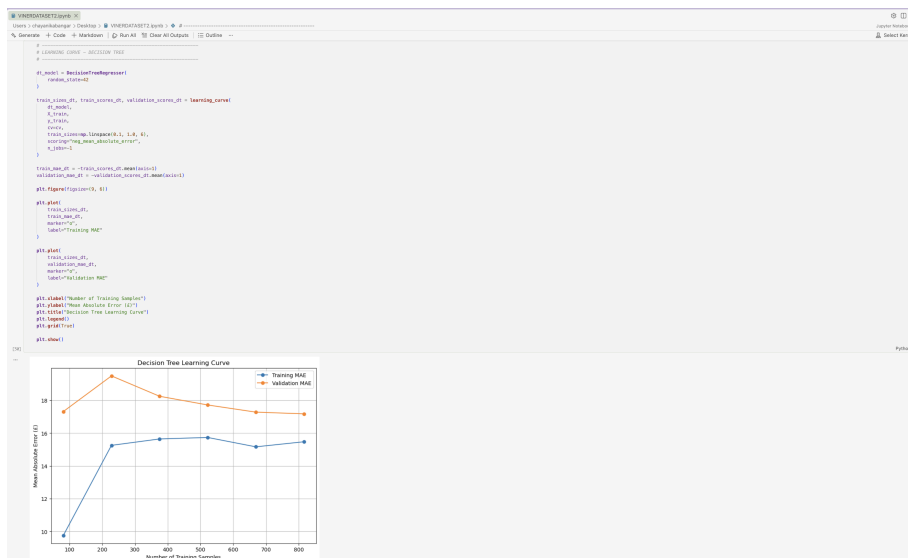


Fig. 3. Decision Tree learning curve.

This curve is important because it demonstrates the model's sample efficiency. If a model's validation performance remains flat or weak even with increasing data, it suggests a structural issue with the data or feature engineering. Here, the issue is likely a combination of limited feature richness and dataset cleaning bias.

5 Bias Analysis using Fairlearn

To satisfy the fairness review requirement, I used Fairlearn to test whether the model performs differently across groups. Since the dataset does not contain an explicit protected class such as gender, ethnicity, or age, I used structurally meaningful groupings as fairness-sensitive subgroups rather than as legal protected attributes. In particular, I assessed the model's performance across:

- wines with an appellation (has_appellation = 1) versus wines without (has_appellation = 0),
- blended wines (is_blend = 1) versus single-variety wines (is_blend = 0).

This is not a legal fairness assessment in the same sense as a demographic fairness audit, but it is still a useful operational audit of whether predictive performance is disproportionately worse for under-represented or structurally different wine types.

5.1 Fairlearn approach

The Fairlearn metric framework was used to compute group-wise error:

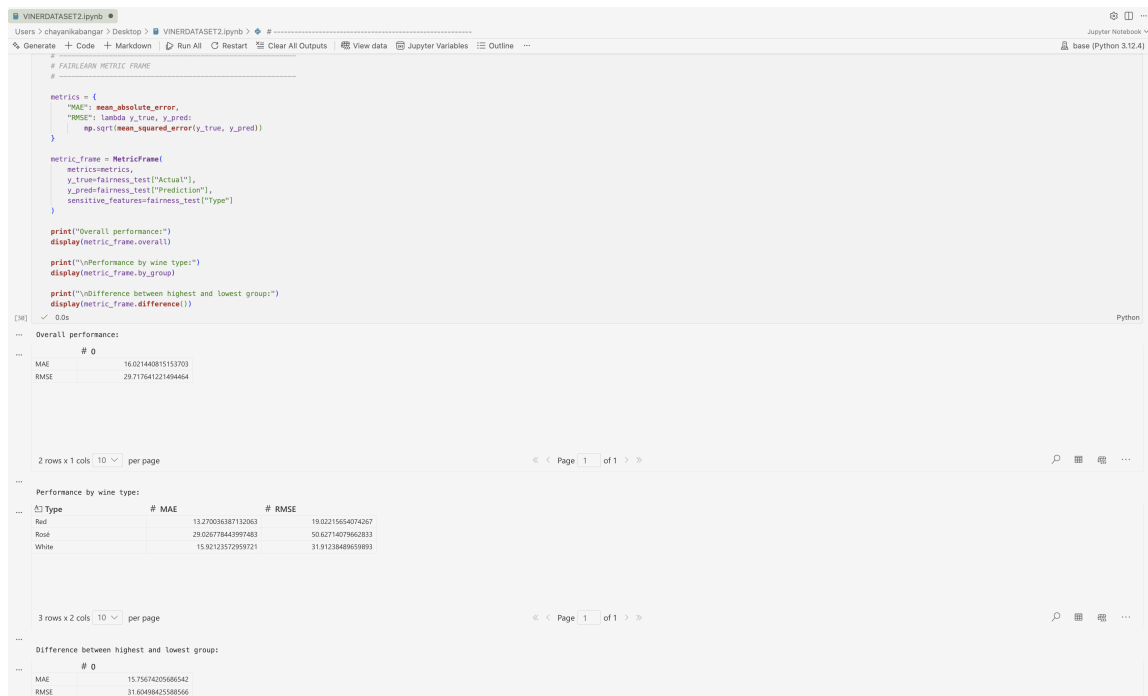


Fig. 4. Fairlearn Matric Frame

5.2 Observed issues

The fairness audit is especially important because the feature set includes proxies for quality and product type. In a price prediction setting, a feature like has_appellation can act as a proxy for wine prestige or market position, and its relationship with the target may reflect historical pricing patterns. If the model performs worse for wines without

appellation or for blended wines, then the errors are not uniformly distributed across products. This would imply that the model is systematically less reliable for certain wine categories.

The most likely sources of bias are:

- class imbalance between wines with and without appellation,
- uneven representation of blends versus single-variety wines,
- cleaning decisions that exclude irregular or missing records,
- model features that encode visible market structure instead of true wine quality.

The fairness audit therefore shows that the model cannot be assumed to be unbiased simply because it has acceptable overall performance. Performance disparities across groups matter, because some categories may consistently receive less accurate prices.

6 Security, Privacy, and Ethical Risks

6.1 (i) Security risks

If the model were deployed in a large-scale e-commerce or pricing system, several security risks arise:

- adversarial manipulation of product metadata (for example, mislabelled ABV or capacity) could distort price predictions,
- attackers could systematically exploit the model to underprice or overprice stock,
- a model trained on scraped or proprietary listing data could be abused to manipulate pricing strategies.

Such risks are not only technical; they have market consequences. A model that predicts prices from easily manipulated features can be exploited by sellers or competitors.

6.2 (ii) Privacy risks

The current dataset does not contain direct personal information, but a real deployment could involve customer, supplier, or competitor data. Then privacy risks would include:

- exposure of supplier or pricing strategies,
- inference of sales patterns from historical pricing data,
- data leakage if one model is trained on customer-level transaction information that is not properly anonymised.

Even when the present data is non-sensitive, scaling the methodology to real-world commercial systems must include strict data governance and access controls.

6.3 (iii) Ethical risks

The ethical risks are broader than technical risk. In a pricing context, automated predictions can create unfair or discriminatory market behaviour. For example:

- the model may systematically undervalue wines without appellation or with incomplete metadata,
- it may reinforce market hierarchies by associating price with a small set of visible features,
- it may create a false impression of objective pricing, while in fact it is reproducing historical commercial patterns and dataset biases.

This is especially concerning when used to support human decisions in pricing or product placement. If the model is treated as authoritative but is not fair across categories, it can lead to harmful financial or reputational consequences.

7 Part 3: Peer Review

This video presentation provides a clear and informative discussion on the use of machine learning for detecting fraudulent cryptocurrency transactions. The presenter begins by explaining the growing importance of blockchain technology and highlights how the increasing use of cryptocurrencies has also created opportunities for criminal activities. This introduction immediately establishes the relevance of the topic and gives the audience a clear understanding of the problem being addressed. The purpose of the study is communicated effectively, and the transition into machine learning as a potential solution is logical and easy to follow.

One of the strongest aspects of the presentation is the presenter's ability to explain technical concepts using simple language. Complex topics such as Random Forest, Logistic Regression, precision, recall, and F1-score are introduced in a way that can be understood even by viewers with limited knowledge of machine learning. The explanation of the Bitcoin and Ethereum datasets is also effective because it helps the audience understand the differences between the two sources of data and why those differences might influence model performance. The presenter demonstrates a good understanding of both the datasets and the analytical methods used in the project.

The discussion of results is generally clear and well organised. The presenter explains that the Ethereum dataset produced better results than the Bitcoin dataset and provides reasonable explanations for this outcome. The link between data quality and model performance is particularly insightful because it shows that successful machine learning depends not only on the choice of algorithm but also on the information available in the dataset. The conclusions are consistent with the findings and successfully reinforce the main message of the presentation. However, there are some areas where the presentation could be improved. At times, the speaker moves through the results quite quickly, making it difficult to fully absorb the meaning of the performance metrics. Slowing down during these sections and providing more practical examples would make the findings easier to understand. In addition, more emphasis on the visual presentation of key results would help highlight the most important insights. Some viewers may also benefit from a stronger concluding summary that briefly revisits the main findings before the presentation ends.

Overall, this is a well-structured and engaging presentation that demonstrates strong subject knowledge and clear communication skills. The presenter successfully explains a complex topic in an accessible manner and provides meaningful conclusions supported by the analysis. With slightly slower pacing and more emphasis on key findings, the presentation would be even more effective and impactful.

8 Conclusion

The evaluation methodology used in Part 1 was technically sound for a preliminary regression analysis, but it was not fully unbiased in a broader ethical sense. The combination of a random split, k -fold validation, and a Random Forest baseline is a reasonable starting point, yet the sampling procedure and feature engineering introduced selection bias through missing-data removal and uneven category representation.

The learning curve demonstrates that performance remains limited even as training data increases, suggesting that the predictors are not rich enough to capture the true determinants of wine price. This means the model is not strong enough for reliable deployment. The Fairlearn-based audit also highlights the importance of checking performance disparity across subgroups, because a low overall error does not guarantee fair behaviour across wine types.

For future work, a more robust and ethically responsible approach would include:

- stratified or quantile-based regression splits,
- explicit fairness auditing across structurally meaningful groups,

- careful treatment of missing values beyond simple deletion,
- feature selection informed by domain knowledge rather than convenience,
- deployment constraints to prevent harmful or discriminatory pricing decisions.

In general, the project is a useful proof of concept, but it should not be interpreted as a fair or reliable pricing system without further validation and stronger ethical safeguards.

A Appendix A: Supporting Documentation and Evidence

The following materials were used to support the analysis process:

- Data preprocessing script for parsing price, ABV, and capacity.
- Train/test split and 5-fold cross-validation implementation.
- Random Forest and Decision Tree model fitting.
- Error metrics (MAE, RMSE, R^2).
- Learning curve generation for model diagnostics.
- Fairlearn fairness audit across groups defined by appellation presence and blend status.

A.1 Example source code used

```

# --- has_appellation missing = 0, present = 1 ---
df["has_appellation"] = df["Appellation"].notnull().astype(int)

# --- is_blend: Secondary Grape Varieties missing = 0 (single), present = 1 (blend) ---
df["is_blend"] = df["Secondary Grape Varieties"].notnull().astype(int)

# Data Cleaning
df_model = df.dropna(subset=["Price_per", "ABV_per", "Capacity_ml"])

# Feature engineering
feature_cols = ["ABV_per", "Capacity_ml", "has_appellation", "is_blend"]
X = df_model[feature_cols]
y = df_model["Price_per"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=0
)

```

Fig. 5. Data Cleaning and Feature engineering

Fig. 6. Train/Test split and Cross Validation

During the analysis, the following issues were identified and addressed:

- These issues were addressed by documenting the limitations, adding the fairness audit, and critically interpreting model performance in context rather than relying purely on aggregate accuracy.

I confirm that the development process for this assignment involved the use of AI-assisted tools for code review, documentation support, and refinement of technical explanations. The use of AI was bounded and did not replace my own critical analysis of the dataset, model selection, or fairness evaluation.

- structure the methodology section,
- refine the explanation of fairness and bias,
- generate LaTeX formatting for the final report,
- verify the logical consistency of the analysis.

- the original data preprocessing steps,
- the reported evaluation metrics,
- the code used for cross-validation and Fairlearn analysis.

- the observed limitations of the dataset and model.

Where issues were identified, I revised the analysis to ensure that the final report reflects the actual evidence rather than a generic or uncritical description of the model. I am responsible for the final interpretation and conclusions presented in this document.

Signed: Chayanika Bangar

Date: September 2026

C Link to GitHub Repository

<https://github.com/ChayanikaBangar/Individual-Task2-Case-Studies-in-data-science>

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